# Required to PENALIZE small values of Precision and Recall

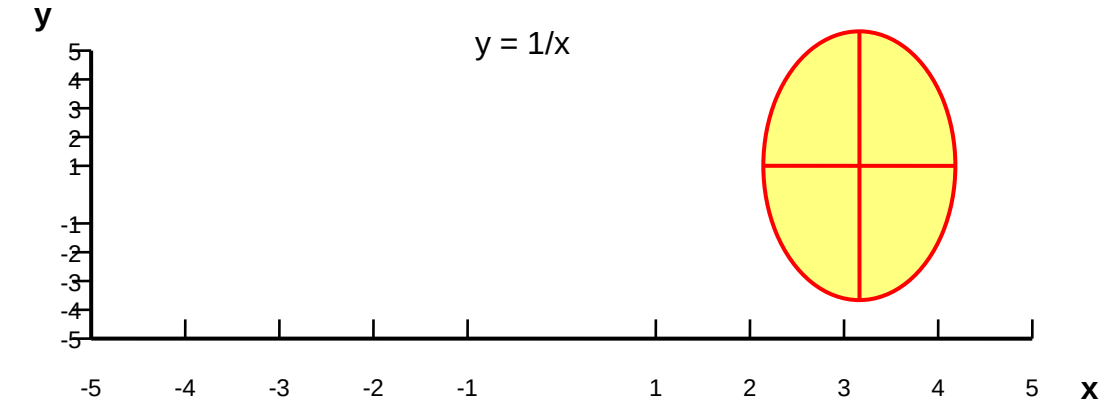
**Remember:**

Ranges of Precision and Recall are from 0 to 1

Using 1/Precision and 1/Recall

As Precision/Recall Decreases ⇒ Value of (1/Precision) and (1/Recall) Increases

Define: 1/F1 = 1/Precision + 1/Recall ⇒ F1 increases as both Precision and Recall are high values




As **Precision/Recall** Decreases ⇒ Value of (1/Precision) and (1/Recall) Increases

Define: **1/F1** = 1/Precision + 1/Recall ⇒ **F1** increases as both **Precision** and **Recall** are high values

F1 Range is from 0 to 0.5 ⇒ **Normalize**

**1/F1 = 0.5 (1/Precision + 1/Recall)**